To Run or Not to Run

Using Pre-Snap Behavior to Predict Play Types in the NFL

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1. Overview/ Introduction:

For our project we decided to participate in the National Football League's Big Data Bowl competition for 2025. The purpose of this competition is to use player and team behavior right before the play begins to gain insight into the resulting play. We realized that the most important information a defense can have is whether the offense intends to execute a run or pass play. Knowing this information would allow the defensive team to move their players around to optimize their chances of getting a tackle or otherwise disrupting the play. We therefore decided to develop statistical models that can predict what kind of play the offense will execute based on game characteristics, team characteristics, and player behavior before and during the snap which we will explore in greater detail later on.

2. Data Sets:

The NFL provided several useful data sets for us to draw from for our analysis.

-games.csv: This set contains all of the game data for the first 9 weeks of the 2022 season.

-plays.csv: This set contains all of the play by play data.

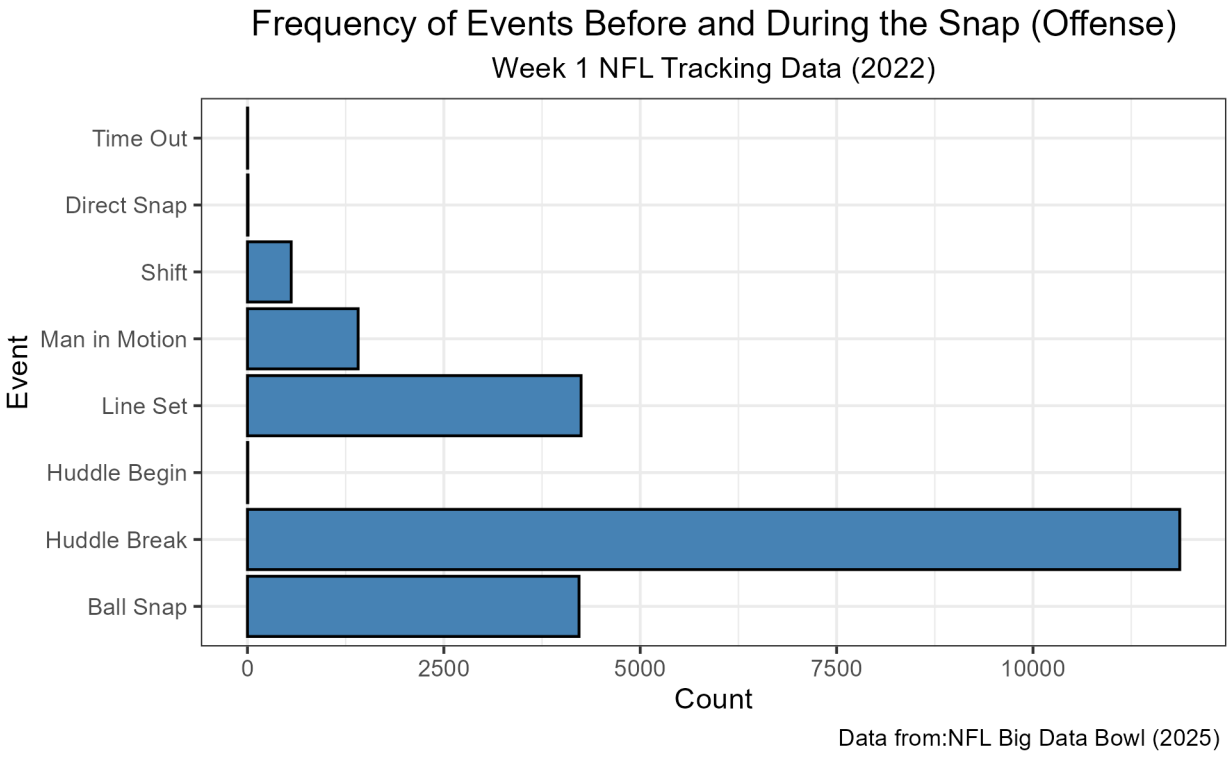
-players.csv: This set contains all of the individual performance data for NFL players.

-player_play.csv: This set contains all of the player level stats for each game and play.

-tracking_week_[week].csv: This set contains all of the tracking data for players and the ball for each game and play in a given week.

3. Exploratory Data Analysis:

Since we were interested to see if things like player motion before or during the snap could be useful in predicting the play type we decided to look at a week of tracking data which contains key pre-snap events like when the offensive line is set (line_set), when the ball is snapped (ball_snap), and whether or not a player was in motion before the snap (man_in_motion) and plot the frequency of these events.


Frequency of Events Before and During the Snap (Offense)
Week 1 NFL Tracking Data (2022)
Time Out
Direct Snap
Shift
Man in Motion
Line Set
Huddle Begin
Huddle Break
Ball Snap
Event
0
2500
5000
7500
10000
Count
Data from:NFL Big Data Bowl (2025)